

ECE113: Basic Electronics (BE)
Winter 2025

End Semester Exam
 Duration: 2 Hours

Date: 30-April-2025

Total Marks: 60

Note: Please go through all questions question before you start solving them.

[CO5] Q1: [5 Marks] A voltage $10 \sin \omega t$ Volt is applied across AB in Fig.-1. Plot the voltage across CD assuming ideal diodes.

[CO3] Q2: [5 Marks] Find the AC voltage V in the circuit shown in Fig.-2 assuming steady state.

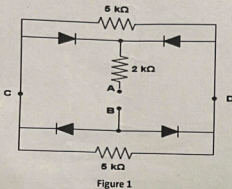


Figure 1

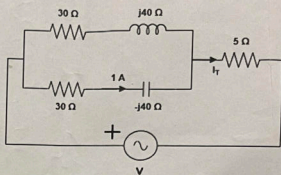


Figure 2

[CO2] Q3: [6 Marks] Find the Norton current and resistance across the Network in Fig.-3 in steady state. Compute the network load impedance such that maximum average power is delivered to the network.

[CO2] Q4: [6 Marks] Find the Thevenin voltage and resistance across the Network and also the value of maximum power delivered to the Network in Fig.-4.

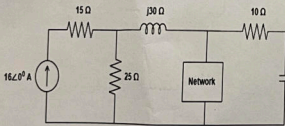


Figure 3

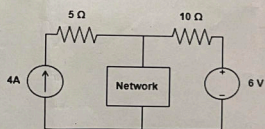


Figure 4

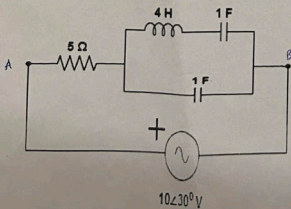
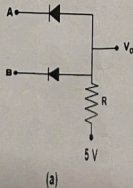
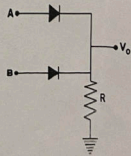


Figure 5



(a)



(b)

Figure 6

[CO3] Q5: [10 Marks] In Fig.-5, assuming steady state determine the AC source frequency such that the average power delivered to the load is maximized and also compute the impedance of the circuit at that frequency between A and B.

[CO5] Q6: [6 Marks] Find the output (V_o) for all possible condition of inputs (A & B) in Fig.-6(a) & (b), where digital input "0" represent "0V" and "1" represent "5V".

[CO1, CO2] Q7: [6 Marks] Find Thevenin's voltage, Thevenin's equivalent resistance at terminal P & Q and voltage V_b in Fig.-7.

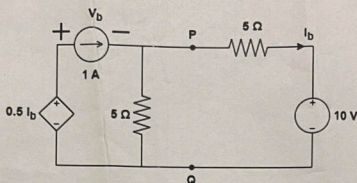


Figure 7

[CO1, CO2] Q8: [10 Marks] In Fig.-8, at time $t=0$ sec the switch S_1 is closed & the switch S_2 is opened. Find the expression of voltage " V " [given that $v_c(0^+) = 2$ volt and $i_L(0^-) = -\frac{1}{6}$ A]. [Hint: You have to find the homogeneous second-order differential equation either in terms of capacitor voltage v_c or inductor current i_L]

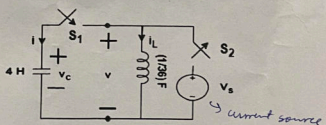


Figure 8

[CO1, CO2] Q9: [6 Marks] In Fig.-9, if the source voltage $V_s = 100 \angle 53.13^\circ$ Volt then find Thevenin's equivalent voltage in volts as seen by the load resistance R_L in steady state.

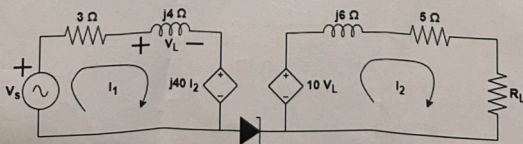


Figure 9